Miter Bend Calculation Report

ASME B31.3-2018 | Internal pressure

DESIGN EQUATION

ASME B31.3-2018, 304.2.3
q = T - c; r2 = (D - T)/2; theta = angle/(2 joints)
(4a) Pm = S E W q/r2 * q/[q + 0.643 tan(theta) sqrt(r2 q)]
(4b) Pm = S E W q/r2 * (R1-r2)/(R1-0.5 r2)
Multiple: min(4a,4b), theta <= 22.5 deg
Single: (4a) if theta <= 22.5 deg; otherwise (4c)
(4c): use 1.25 instead of 0.643 in (4a)
(5) R1 >= A/tan(theta) + D/2 (empirical A, not allowance c)
Pressure equations: P and S in MPa; dimensions in mm; input bar x 0.1 = MPa.

Report date	**2026-10-06 10:39**

01 INPUTS

Design gauge pressure	**20 bar**
Miter cut angle theta	**22.5 deg**
Material / product	**ASTM A106 Gr.B (Carbon Steel, Seamless)**
Design metal temperature	**20 C**
S at design temperature	**138 MPa; user supplied, A-1/A-1M**
E / W	**1 / 1**
OD D / segment wall T	**114.3 / 6.02 mm**
Total allowances c	**1.5 mm**
Miter weld joint count	**2**
Effective radius R1	**150 mm**

02 RESULTS

Governing equation	**Eq. 4a & 4b (multiple miter)**
Maximum pressure Pm	**59.951 bar**
Pressure margin Pm-P	**39.951 bar**
Required R1 / wall extension M	**117.5053 / 45.1333 mm**
Minimum T meeting pressure	**3.4453 mm; numerical solution**

MEETS the implemented internal-pressure equations.

Provide the required extension M in the actual fabrication geometry; the app does not measure it. Overall design, material suitability, fabrication and other loads require independent verification.

04 REFERENCES AND LIMITS

Sources: B31.3-2018 304.2.3, supplied printed pp.22-23; B31.1-2022 104.2.3, supplied printed p.29. Inputs are user supplied. No external-pressure, fatigue, remaining-life or complete code qualification is performed.